

Problem 3

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Problem 1 Assume that the rate of change (in dollars per day) of the price of shares of stock in the WeSaySo Company (with t in days) is modeled by the equation $r(t) = -3t^2 + 30t - 63$ (note that this is technically a discrete function, but prices change so often with stocks that modeling this with a continuous function makes sense). Assume also that the price of a share of stock on day 1 (i.e., $t = 1$) is \$51. Answer the following questions:

- (a) Find the rate of change of price at $t = 5$.
- (b) Find the price of a share of stock at $t = 5$.
- (c) How fast is the rate of change of price changing at $t = 5$?
- (d) How much did the price of a share of stock change in the first 6 days (i.e., on $[1, 6]$)?
- (e) What was the greatest rate of change of price during the first 6 days (i.e., on $[1, 6]$)? [3]
- (f) What was the greatest price of a share of stock during the first 6 days (i.e., on $[1, 6]$)? [3]